GROUP - 66 Simple Multithreader Documentation

Steps to run on terminal:

1. Path of the file
2. Make
3. ./vector <numThreads> <size of vector>
   ./matrix <numThreads> <matrix dimension>
4. Execution time and success message will be displayed

Implementation:

- part_size() splits the total elements into <numThreads> parts
  In case of uneven split, the first <extra> threads get 1 extra element
- split_range() splits the range among <numThreads> threads using part_size() to determine the sub-ranges
- The structs thread1D, thread2D contain the the range for that thread and its corresponding lambda function
- create_and_join() (handles both 1D and 2D cases) - creates <numThreads-1> helper threads and waits for each thread to finish
  The main thread executes the last part
- start1D(), start2D() call corresponding lambda functions for each element
- std::chrono::steady_clock::now() for execution time
- parallel_for
  1. Split range
  2. Assign each part to thread
  3. Create <numThreads-1> helper threads
  4. Main thread executes last part
  5. All threads concurrently run the lambda
  6. Wait for all threads to finish
  7. Print execution time

- Proper error handling implemented for invalid numThreads, pthread_create, pthread_join

Github repository link: https://github.com/antelope0112/group66-multithreader